Paper 1 Updated Report

Last updated: 2026-11-10

Deadline: 2026-11-12. This replaces the Paper 1 section of Paper Readiness Check. It is based on real training runs: 280 runs on a Colab T4 on 2026-11-10, all in `paper1_results/results/runs.jsonl` in the public repository github.com/AakashBhat1/zeroday-intrusion-detection-ssl. The compiled PDF still shows the old numbers; the LaTeX source has not been provided, so no text has changed yet.

Paper 1: `1/FINAL_ICIDA2026_SUBMISSION_PHASE25.pdf`

Self-supervised tabular model for zero-day detection on CSE-CIC-IDS2018. Six models, four label budgets, ten seeds, plus an objective ablation and a shortcut check.

Headline at the 10% budget: decoupled zero-day AUROC 0.9785 ± 0.0028 (printed: 0.9343 ± 0.0420), Bot recall 0.9827 ± 0.0037 at a 0.0436 false-positive rate (printed: 0.3763 ± 0.3918), closed-world Macro-F1 0.7001 ± 0.0234 (printed: 0.5240 ± 0.0216).

Resolved from the readiness check: - Leakage: the pretraining pool is now the training split only (160,526 flows). Eleven leakage checks pass, including no shared rows with validation and no feature-identical flows between historical data and the test set. - Epochs: 10 pretraining epochs, 5 warm-up, centroid fixed after epoch 5. Tuning showed 30 epochs is no better. - Table 5 statistics: recomputed with an exact Wilcoxon test. Raw p is 0.0020 or 0.0039, adjusted 0.0029 or 0.0039; all six comparisons are significant. - Parameter count: the components match the paper (316,424 / 102,024 / 49,984 / 9,294). The trained total is 477,726, not 511,006. - `L_cons`: rows D and E are dropped. The ablation is now A, B, C, F over 10 seeds each. - Decoupled vs MSP: both scores now come from one probe, so the comparison is a clean scoring ablation. - Deep SVDD threshold: the 1e11 scale is gone (0.019 ± 0.002) after clipping scaled features to ±10.

Must change in the text: - Every number in the abstract, Tables 2-5, Figs. 1-3 and sections 5-7. - Flow count: 16,232,943 flows (16,233,002 counts 59 repeated header lines). - Split sizes: 160,526 / 20,066 / 20,066 / 100,000; budgets 16,052 / 8,025 / 1,606. - Quarantine: "at least 12 hours". The source timestamps use a 12-hour clock, so "12 hours, 6 seconds" is an artefact. - Shortcut paragraph: remove or rewrite. Random Forest scores 0.9523 with port and protocol and 0.9528 without. - "Mixture calibration collapses recall": no longer true (recall 0.9630 ± 0.0538). - "Deep SVDD achieves superior ranking": no longer true at 10% (0.7993). It is true at 1% (0.9833). - Table 2 caption: Tesla T4 and 2 vCPU, not RTX 3050. - Feature-identical validation flows: 8,881, not 1,606. State that 32,929 test-day flows repeating a historical feature vector were removed. - Reference 8 title: "Pre-training Transformers".

Claims to soften: - Not best at every budget. Random Forest ties at 100% labels (0.9779 vs 0.9785); Deep SVDD wins at 1%. - "Across all budgets" is constant by construction: the latent score uses no labels. - Trees lead closed-world classification at every budget (Macro-F1 0.83-0.85 vs 0.70 at 10%). - Bot is a favourable case. On four stand-in zero-days (one known attack held out) the decoupled score averages 0.68 AUROC, against 0.83 for benign-distance scores. This needs a limitations paragraph.

New evidence the paper can use: - Ablation: decoupled AUROC 0.6370 (reconstruction), 0.8077 (contrastive), 0.9216 (both), 0.9785 (with the novelty loss). - Encoder value: the same benign-distance scores on raw features score 0.7434 and 0.4214 on Bot. - Stability: seed-to-seed std of 0.0028, about a tenth of any baseline's.

Table 3 row (10%)	AUROC	AUPRC	CW Macro-F1
S4_E06 (Decoupled)	0.9785 ± 0.0028	0.9423 ± 0.0095	0.7001 ± 0.0234

Table 3 row (10%)	AUROC	AUPRC	CW Macro-F1
S4_E06 (MSP)	0.8747 ± 0.0822	0.7950 ± 0.1136	0.7001 ± 0.0234
S4_E05	0.8818 ± 0.1086	0.8739 ± 0.0896	0.7471 ± 0.0103
Random Forest	0.9528 ± 0.0207	0.9376 ± 0.0261	0.8267 ± 0.0152
XGBoost	0.8352 ± 0.0946	0.7938 ± 0.1217	0.8493 ± 0.0161
Deep SVDD	0.7993 ± 0.0277	0.8152 ± 0.0220	N/A
SCARF	0.4672 ± 0.0933	0.6066 ± 0.1103	0.6869 ± 0.0250

The full set of tables is in `paper1_results/report/` (CSV and LaTeX). Code: `src/zdssl/`. Split sizes and leakage checks: `manifest.json`. Stand-in zero-day runs: `tune.jsonl`. The same tables are also in the vault note Paper 1 Results.

Not done

- Paper text: blocked on the LaTeX source.
- Fig. 3 and the Bot flow profile (the "650 / 49,350" sentence) are not regenerated.
- Five model improvements are planned and not started: section 13 of Paper 1 Training Plan.
- Paper 2 is unchanged from Paper Readiness Check.